

Operating Instructions

Mobile Robot – Border Guard Operations

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Client: Border Guard Officers (Poland and Lithuania)

Document Type: Operating Instructions (ISO-compliant)

Applicable Standards: ISO 12100, ISO 20607, ISO 8373, ISO 10218-1 (robot safety principles)

1. Purpose and Scope

This document provides standardized operating instructions for the deployment of the mobile robot in airport and border-zone security operations.

The robot is intended for:

- Pyrotechnic threat neutralization
- Suspicious object reconnaissance
- Chemical hazard identification
- Rescue and negotiation

NOTE: This document applies to trained Border Guard officers authorized to operate robotic systems in controlled and civilian environments.

2. System Overview

Component	Description
Mobile Platform	6-wheel drive, max speed 10 km/h, terrain-adaptive suspension
Manipulator Arm	Telescopic reach 3.3 m, payload capacity 50 kg
Power Supply	Rechargeable battery, approx. 4 h runtime, hot-swappable
Control Console	Wireless range up to 1000 m (open terrain), ergonomic interface
Optional Attachments	X-ray holder, pyrotechnic launcher, gripper jaws, negotiation set

WARNING: Only certified operators may attach, remove, or configure mission-specific modules.

3. Safety Requirements

3.1 General Safety Rules

WARNING – RISK OF SERIOUS INJURY: Improper robot operation may endanger personnel and civilians.

Operators must:

- Maintain a minimum 5-meter safety distance from the robot during movement.
- Keep civilians outside the operational perimeter.
- Follow all command-center instructions.
- Use PPE appropriate to the mission (ballistic vest, gloves, hearing protection).

3.2 Environmental Safety

- Ensure terrain is suitable for robot mobility.
- Avoid waterlogged, unstable, or explosive-risk surfaces.
- Maintain line-of-sight when possible.

3.3 Equipment Safety

- Verify attachment integrity before each mission.
- Do not exceed manipulator payload limits.
- Do not operate the robot near flammable vapors or open flames.

4. Pre-Operation Checklist

Before deployment, confirm:

- Battery fully charged; spare battery available
- Manipulator arm calibrated; joints tested
- Wireless connection established with control console
- Cameras and sensors operational
- Attachments securely mounted
- Emergency stop and override systems functional

NOTE: Any failed checklist item must be reported before mission start.

5. Basic Operation

5.1 Startup Sequence

1. Power on the robot using the rear panel switch.
2. Launch the control console software.
3. Connect to the robot (IBIS® interface).
4. Confirm telemetry: battery, signal strength, motor status.

5.2 Navigation

- Use joystick or touchscreen controls.
- Monitor terrain via onboard cameras.
- Adjust speed according to surface conditions and threat proximity.
- Avoid sudden turns when carrying payload.

5.3 Manipulator Operation

1. Select object-handling mode.
2. Extend arm using telescopic controls.
3. Grip object using appropriate jaw type.
4. Lift slightly to confirm grip stability.
5. Retract arm and reposition as required.

6. Mission Scenarios

Scenario	Recommended Configuration
Suspicious Package Inspection	X-ray holder + manipulator arm
Explosive Neutralization	Pyrotechnic launcher + remote detonation protocol
Chemical Threat Detection	Sensor module + sealed transport container
Hostage/Negotiation Support	Negotiation set + audio relay

7. Emergency Protocols

Scenario	Recommended Configuration
Signal Loss	Robot enters safe mode; operator must initiate manual override.
Obstacle Collision	Auto-stop activates; inspect robot before continuing.
Manipulator Overload	System alert triggers; reduce payload immediately.
Battery Depletion	Return to base or replace battery module.

WARNING: Do not approach the robot until motors are fully powered down.

8. Post-Operation Procedures

- Power down robot and console.
- Clean surfaces; inspect for mechanical wear.
- Log mission data and maintenance notes.
- Recharge battery and store robot in secure location.

9. Maintenance Schedule

Task	Frequency
Battery health check	Weekly
Joint lubrication	Monthly
Software updates	As released
Sensor calibration	Before each mission
Attachment inspection	After each use

10. Operational Protocol

Relocation of Unsupervised Suitcase – Airport Zone

Scenario: Suspicious luggage handling in civilian airport environment

Robot Model: Pyrotechnic and Reconnaissance Platform

10.1 Objective

To safely relocate an unsupervised suitcase from a public airport zone to a designated containment area, minimizing risk to personnel and civilians.

10.2 Safety Precautions

Before operation:

- Evacuate civilians within a 30-meter radius
- Establish secure communication with command center
- Confirm readiness of emergency response units
- Assume potential presence of explosives or hazardous materials
- Verify robot functionality and attachment integrity

WARNING: Treat all unattended luggage as potentially explosive.

10.3 Equipment Configuration

Component	Configuration Notes
Manipulator Arm	Telescopic mode, soft-pad gripper jaws
X-ray Holder	Mounted if preliminary scan required
Control Console	Wireless mode, operator in secure zone
Sensor Module	Activated for chemical/explosive trace detection

10.4 Operational Steps

10.4.1 Deployment

1. Transport robot to perimeter of incident zone.
2. Power on and establish console connection.
3. Activate cameras and sensors.

10.4.2 Approach

1. Navigate slowly toward suitcase.
2. Maintain visual feed and telemetry monitoring.
3. Stop at 1 meter distance for preliminary scan (if equipped).

10.4.3 Object Handling

1. Extend manipulator arm.
2. Secure suitcase handle or frame with gripper.
3. Lift slightly to confirm grip.
4. Retract arm and retreat slowly.

10.4.4 Relocation

1. Follow pre-cleared path to containment area.
2. Place suitcase in secure enclosure.
3. Release grip and retract arm.

11. Post-Operation Protocol

- Notify command center of successful relocation.
- Inspect manipulator for wear.
- Log operation details (object weight, handling time, sensor readings).
- Submit video and telemetry for review.

12. Incident Documentation

Field	Example Entry
Date/Time	2015.10.11, 12:45 CET
Location	Warsaw Chopin Airport, Terminal A
Object Description	Medium-sized black suitcase, un-tagged
Robot Operator	BG Officer Kowalski
Outcome	Object relocated, no threat detected

13. Console Operation Guide (Relocation Scenario)

13.1 Console Startup

1. Power on console.
2. Launch IBIS® interface.
3. Select *Mission Mode: Recon & Relocation*.
4. Confirm wireless connection (status: green).
5. Verify camera feed and telemetry.

13.2 Navigation Controls

Function	Input
Forward/Reverse	Left joystick – vertical
Turn Left/Right	Left joystick – horizontal
Speed Adjustment	Low/Medium/High toggle
Camera View	Front/Arm/Rear selector
Zoom/Focus	Touchscreen or dial

TIP: Use low-speed mode when approaching the suitcase.

13.3 Manipulator Controls

Function	Input
Arm Extend/Retract	Right joystick – vertical
Arm Elevation	Right joystick – horizontal
Gripper Open/Close	Gripper toggle
Grip Pressure	Touchscreen slider
Lift/Lower	Arm lift dial

Procedure:

1. Switch to Manipulator View.
2. Extend arm toward suitcase.
3. Secure object gently.
4. Confirm grip stability.
5. Lift slightly and retract.

13.4 Relocation Sequence

1. Switch to Rear View.
2. Navigate to containment zone.
3. Monitor object stability.
4. Position over containment platform.
5. Release grip and retract arm.
6. Confirm placement via camera feed.

13.5 Emergency Console Functions

Situation	Action
Signal Loss	Activate Safe Mode override
Manipulator Overload	Reduce grip pressure or abort
Obstacle Detected	Engage Auto-Stop
Battery Warning	Initiate Return-to-Base

13.6 Logging and Shutdown

1. Save mission log (video + telemetry).
2. Enter incident metadata.
3. Power down robot and console.
4. Submit report to command center.

14. Operator Experience Summary

- Controls are intuitive and easy to learn.
- Initial missions may feel stressful due to equipment value.
- Confidence increases with practice.
- High-quality camera views improve precision.
- Mission logs support skill development